K.R. MANGALAM UNIVERSITY

Unit–4 Assignment: Non-Linear and Advanced Data Structures

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COURSE - Data Structures

PROGRAMME - B. Tech CSE (AI & ML)

SECTION - A

SUBMITTED TO - Mrs. Neetu Chauhan

Task 1: Binary Search Tree (BST)

A Binary Search Tree (BST) is a hierarchical data structure where:

- Each node has at most two children.
- The left subtree contains values smaller than the node.
- The right subtree contains values greater than the node.

This structure allows efficient searching, insertion, and deletion.

Why Inorder Traversal Gives Sorted Output ?

In inorder traversal, nodes are visited in this order:
Left → Root → Right

- First, all smaller elements (left subtree) are visited.
- Then the current node.
- Then all larger elements (right subtree).

Because of the BST property (left < root < right), this traversal naturally produces elements in ascending sorted order.

<u>Time Complexity</u>

- **<u>Average Case:</u>**
  Operations like search, insert, and delete take **O(log n)** time because the tree is balanced and we eliminate half of the nodes at each step.
- **<u>Worst Case:</u>**
  Time becomes **O(n)** when the tree becomes skewed (like a linked list), for example when elements are inserted in sorted order.

Task 2: Graph Representation, BFS and DFS

A graph is a collection of nodes (vertices) connected by edges.
In this assignment, the graph is:

- Directed (edges have direction)
- Weighted (each edge has a cost)
- Stored using an adjacency list

Example:

A → [(B,2), (C,4)]

BFS vs DFS (Simple Difference)

BFS (Breadth-First Search)	**DFS (Depth-First Search)**
Visits nodes level by level	Goes deep into one branch first
Uses a queue	Uses recursion or stack
Finds shortest path (unweighted graph)	Explores full paths

Real-World Use Cases

- **BFS:** Used in finding the shortest path in navigation systems (like maps, GPS).
- **DFS:** Used in detecting cycles in graphs (important in compilers and networking).

Task 3: Hash Table with Separate Chaining

A hash table stores data in key-value pairs for fast access.
A hash function is used to convert a key into an index:

index = key % table_size

1.What is a Collision?

A collision occurs when:

- Two or more keys produce the same index.

Example:

10 % 5 = 0
15 % 5 = 0
20 % 5 = 0

All these keys map to the same index → collision occurs.

2.Why Separate Chaining Helps?

In separate chaining:

- Each index stores a list (bucket) of elements.
- Multiple values can be stored at the same index.

So instead of overwriting data, we store all collided elements safely.

<u>Time Complexity</u>

- **<u>Average Case:</u>**
  O(1) → Very fast lookup when collisions are few.

- **Worst Case:**
  O(n) → When many elements fall into the same bucket.

Conclusion

In this assignment, I implemented and studied three important data structures: Binary Search Tree, Graph, and Hash Table. The BST helped in storing data in a sorted and efficient manner with operations like insertion, search, deletion, and inorder traversal. The Graph module demonstrated how BFS and DFS are used to explore networks in different ways. The Hash Table showed how key-value pairs can be stored and accessed quickly while handling collisions using separate chaining. Overall, this assignment improved my understanding of how these data structures work and their real-world applications in systems like databases, networks, and caching.

References

- GeeksforGeeks
- W3Schools
- Notes from the lecture